Worked Solutions to Chapter 4: Fractals

Chaos: An Introduction to Dynamical Systems (Alligood, Sauer, Yorke)

Ali Raza Jatoi

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This document solves the end-of-chapter Exercises 4.1–4.14 and Challenge 4 of Chapter 4. Each solution starts with the main idea, then gives the steps, and ends with a boxed answer. Problem statements are paraphrased, so keep the book open next to this text. Final answers were checked against the book's "Answers and Hints" appendix wherever it gives one.

Toolkit: facts used throughout

Toolkit

1. **Box-counting dimension.** Let $N(\varepsilon)$ be the number of boxes of side ε (from a fixed grid) that meet the set S. Then $\operatorname{boxdim}(S) = \lim_{\varepsilon \to 0} \dfrac{\ln N(\varepsilon)}{\ln(1/\varepsilon)}$ when the limit exists.
2. **Counting shortcut.** If at scale $\varepsilon_n = c\, r^{-n}$ we have $N(\varepsilon_n)$ within constant factors of m^n, then $\operatorname{boxdim}(S) = \dfrac{\ln m}{\ln r}$. Constant factors drop out because $\ln C/(n \ln r) \to 0$. Since $N(\varepsilon)$ only grows as ε shrinks, scales between ε_{n+1} and ε_n give the same limit.
3. **Middle-third Cantor set K.** $K = \bigcap_n K_n$, where K_n has 2^n closed intervals of length 3^{-n}. Equivalently, K is the set of $x \in [0,1]$ with a ternary expansion using only the digits 0 and 2. Endpoints of every interval of K_n stay in K forever. $\operatorname{boxdim}(K) = \ln 2/\ln 3$.
4. **Measure zero.** S has measure zero if, for every $\varepsilon > 0$, countably many intervals of total length $\leq \varepsilon$ cover S.
5. **Countable** means "can be listed $x_1, x_2, x_3, \ldots$". $\mathbb{Q}$ is countable and $[0,1]$ is not.

Exercises

Exercise 4.1

(a) Match the part of K inside $[0, \frac{1}{3}]$ one-to-one with all of K. (b) Repeat with $[\frac{19}{27}, \frac{20}{27}]$.

(a) Points of K have ternary expansions with digits $0, 2$ only. A point of K lies in $[0, \frac{1}{3}]$ exactly when its first digit is 0, i.e. $x = 0.0a_2a_3a_4\ldots_3$ with $a_i \in \{0, 2\}$. Define $f(x) = 3x$. Multiplying by

3 shifts the digits left, so $f(x) = 0.a_2a_3a_4\ldots_3 \in K$. Conversely, for $y = 0.b_1b_2\ldots_3 \in K$, the point $y/3 = 0.0b_1b_2\ldots_3$ lies in $K \cap [0, \frac{1}{3}]$. So f is a one-to-one map of $K \cap [0, \frac{1}{3}]$ *onto* K (its inverse is $y \mapsto y/3$).

(b) A check first: $\frac{19}{27} = 0.201_3$ and $\frac{20}{27} = 0.202_3$, and $(\frac{19}{27}, \frac{20}{27})$ is one of the removed middle thirds at stage 3. So $K \cap [\frac{19}{27}, \frac{20}{27}]$ is only two points, and no one-to-one correspondence with all of K is possible. This looks like a misprint in the book. The idea of (a) works for *any* interval of K_n. For example, for $[\frac{20}{27}, \frac{21}{27}]$ use $f(x) = 27x - 20$ (inverse $y \mapsto (20 + y)/27$), and for $[\frac{18}{27}, \frac{19}{27}]$ use $f(x) = 27x - 18$. In general, an interval $[a, a + 3^{-n}]$ of K_n is matched with K by $f(x) = 3^n(x - a)$.

Answer

(a) $f(x) = 3x$ is a bijection from $K \cap [0, \frac{1}{3}]$ onto K. (b) The same works with $f(x) = 3^n(x - a)$ on any interval $[a, a + 3^{-n}]$ of K_n. The interval $[\frac{19}{27}, \frac{20}{27}]$ as printed is a removed gap, so it is probably a misprint for $[\frac{20}{27}, \frac{21}{27}]$ or $[\frac{18}{27}, \frac{19}{27}]$.

Exercise 4.2

The middle-half Cantor set $K(4)$: delete the middle half of every subinterval. Find the removed length, the members, whether $\frac{1}{5}$ and $\frac{17}{21}$ belong, the dimension, and the tent-map slope that gives $K(4)$.

Stage 1 removes $(\frac{1}{4}, \frac{3}{4})$ and leaves $[0, \frac{1}{4}] \cup [\frac{3}{4}, 1]$. Stage n leaves 2^n intervals of length 4^{-n}.

(a) The remaining length is $2^n \cdot 4^{-n} = 2^{-n} \to 0$, so the removed length is 1. (Or: $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = 1$.)

(b) The first stage keeps numbers whose first base-4 digit is 0 or 3. The same cut is repeated inside each piece, one digit further along. So $K(4)$ consists of the numbers in $[0, 1]$ whose **base-4 expansion uses only the digits** 0 **and** 3. (Endpoints may need their alternate expansion, e.g. $\frac{1}{4} = 0.1_4 = 0.0333\ldots_4$.)

(c) For $\frac{1}{5}$: $4 \cdot \frac{1}{5} = \frac{4}{5} \Rightarrow$ digit 0; $4 \cdot \frac{4}{5} = \frac{16}{5} = 3 + \frac{1}{5} \Rightarrow$ digit 3; now we are back at $\frac{1}{5}$. So $\frac{1}{5} = 0.030303\ldots_4 \in K(4)$. For $\frac{17}{21}$: $4 \cdot \frac{17}{21} = 3 + \frac{5}{21}$ (digit 3), $4 \cdot \frac{5}{21} = \frac{20}{21}$ (digit 0), $4 \cdot \frac{20}{21} = 3 + \frac{17}{21}$ (digit 3), and we are back at $\frac{17}{21}$. So $\frac{17}{21} = 0.\overline{303}_4$ uses only digits $0, 3$ and *is* in $K(4)$.

(d) Length $= 0$ by (a). Since $N(4^{-n}) = 2^n$, $\operatorname{boxdim} K(4) = \dfrac{\ln 2^n}{\ln 4^n} = \dfrac{\ln 2}{\ln 4} = \dfrac{1}{2}$.

(e) The tent map is $T_a(x) = ax$ for $x \le \frac{1}{2}$ and $a(1 - x)$ for $x \ge \frac{1}{2}$. For $a > 2$, points with $T_a(x) > 1$ escape, namely $x \in (\frac{1}{a}, 1 - \frac{1}{a})$. This removed middle interval has length $1 - \frac{2}{a}$. Setting $1 - \frac{2}{a} = \frac{1}{2}$ gives $a = 4$. For $a = 4$, each surviving interval $[0, \frac{1}{4}]$ and $[\frac{3}{4}, 1]$ is mapped linearly onto $[0, 1]$, so the same cut repeats inside each one. Hence $S = K(4)$.

Answer

(a) 1. (b) Numbers whose base-4 expansion uses only digits $0, 3$. (c) Both $\frac{1}{5} = 0.\overline{03}_4$ and $\frac{17}{21} = 0.\overline{303}_4$ are in $K(4)$. (d) Length 0, boxdim $= \frac{1}{2}$. (e) $a = 4$.

Exercise 4.3

Let T be the tent map with $a = 2$. (a) Rationals $\to$ rationals, irrationals $\to$ irrationals. (b) Periodic points are rational. (c) Uncountably many nonperiodic points.

(a) $T(x) = 2x$ or $2 - 2x$, which are rational whenever x is. Conversely, if $T(x) = q$ is rational then $x = q/2$ or $x = 1 - q/2$ is rational. So an irrational x cannot have a rational image: irrationals map to irrationals.

(b) Each branch of T has the form $\pm 2x +$ integer (namely $2x$ or $-2x + 2$). Composing, on any piece $T^n(x) = \pm 2^n x + c$ with c an integer: if $T^k(x) = \sigma 2^k x + c_k$, then $T^{k+1}(x) = 2(\sigma 2^k x + c_k)$ or $2 - 2(\sigma 2^k x + c_k)$, which again has this form with integer constant. If p has period $n \geq 1$, then $p = \pm 2^n p + c$, so $p = \dfrac{c}{1 \mp 2^n}$. The denominator is nonzero for $n \geq 1$, so p is rational.

(c) By (b) the periodic points lie in $\mathbb{Q}$, which is countable. The irrational numbers of $[0, 1]$ are uncountable and, by (a), none of them is periodic. (More generally, uncountable minus countable is uncountable.)

Answer

Rationals map to rationals and irrationals to irrationals. A period-n point satisfies $p = c/(1 \mp 2^n)$ with $c \in \mathbb{Z}$, hence is rational. The irrationals in $[0, 1]$ give uncountably many nonperiodic points.

Exercise 4.4

Show that the set of rational numbers has measure zero.

The rationals are countable, so list them $q_1, q_2, q_3, \ldots$. Given $\varepsilon > 0$, cover q_n with an interval of length $\varepsilon/2^n$. The total length is at most $\varepsilon \sum_{n \geq 1} 2^{-n} = \varepsilon$. Since ε is arbitrary, $\mathbb{Q}$ has measure zero. $\square$

Answer

Cover the nth rational by an interval of length $\varepsilon/2^n$; the total length is $\leq \varepsilon$ for every $\varepsilon > 0$.

Exercise 4.5

Let $\hat{K} \subset K$ be the points whose ternary expansion does not end in repeating 2's. Show $\hat{K}$ matches $[0, 1]$ one-to-one, so K is uncountable.

Every $x \in K$ has a *unique* expansion $x = \sum a_i 3^{-i}$ with $a_i \in \{0, 2\}$. Define

$$\varphi(x) = \sum_{i \geq 1} \frac{a_i/2}{2^i},$$

i.e. replace each digit 2 by 1 and read the result in base 2.

Injective: different digit strings give different x, and different binary strings that do not end in all 1's give different numbers (each number in $[0, 1)$ has exactly one binary expansion not ending in all 1's).

Onto $[0,1)$*:* given $y \in [0,1)$, take its binary expansion not ending in all 1's, replace each 1 by 2, and read in base 3. The result is in $\hat{K}$ and maps to y.

So $\hat{K} \leftrightarrow [0,1)$. To include the endpoint 1, move it into the set by shifting the sequence $1, \frac{1}{2}, \frac{1}{3}, \ldots$ one step ($\frac{1}{n} \mapsto \frac{1}{n+1}$) and fixing everything else. This gives a bijection $[0,1] \to [0,1)$. Since $[0,1]$ is uncountable and $\hat{K} \subset K$, the set K is uncountable. □

Answer

$\varphi\big(\sum a_i 3^{-i}\big) = \sum (a_i/2)\, 2^{-i}$ is a bijection from $\hat{K}$ onto $[0,1)$ (and, with the one-point shift above, onto $[0,1]$). Hence K is uncountable.

Exercise 4.6

(a) K is closed. (b) K is perfect, hence a Cantor set. (c) Any (bounded) Cantor set can be matched with K one-to-one, preserving order.

(a) $K = \bigcap_n K_n$ and each K_n is a finite union of closed intervals, hence closed. An intersection of closed sets is closed.

(b) Let $p \in K$ and $n \geq 1$. Then p lies in some interval I_n of K_n, of length 3^{-n}. Both endpoints of I_n belong to K, and at least one of them differs from p. That endpoint is within 3^{-n} of p. As n is arbitrary, every neighborhood of p contains a point of K other than p, so p is a limit point of K. With (a), K is perfect. Also K contains no interval: an interval inside $K \subset K_n$ must sit in a single component of K_n (components are separated by gaps), so its length is at most 3^{-n} for every n, hence 0. So K is a Cantor set.

(c) *Assumption:* S is bounded (the book's pictures are; an unbounded S has no maximum, while K has one, so no order-preserving match could exist). Let S be a bounded perfect set with no intervals, and write $a = \min S$, $b = \max S$. We build a "binary address" for every point of S.

Splitting step. Let $P = S \cap [u,v]$ with $u < v$ both in S, and $L = v - u$. The open middle third of $[u,v]$ is not contained in S, so it contains a point $z \notin S$. Let $c = \sup(S \cap [u,z])$ and $d = \inf(S \cap [z,v])$. Then (c,d) is a gap: $c, d \in S$, $c < z < d$. Because S is perfect, u and v are not isolated points, so $u < c$ and $d < v$. Also $c - u < \frac{2}{3}L$ and $v - d < \frac{2}{3}L$. Split P into the left piece $S \cap [u,c]$ (label 0) and the right piece $S \cap [d,v]$ (label 1).

Iterate. Start with $P = S \cap [a,b] = S$ and split repeatedly. After n splits there are 2^n pieces, each of diameter at most $(\frac{2}{3})^n (b-a) \to 0$.

- Every infinite 0/1 string gives nested nonempty compact pieces with diameter $\to 0$, so they intersect in exactly one point of S.
- Every $s \in S$ lies in exactly one piece at each stage (the gaps contain no points of S), so it has exactly one address.
- Different strings give disjoint pieces after the first differing bit, hence different points.

So $S \leftrightarrow \{0,1\}^{\mathbb{N}}$ is a bijection. Doing the same for K (left third = 0, right third = 1) gives $K \leftrightarrow \{0,1\}^{\mathbb{N}}$. Composing, we get a bijection $S \to K$. *Order:* if $s < t$ in S, at the first bit where their addresses differ, s is in the left piece and t in the right piece. The same is true for their images in K, so $s < t \Rightarrow s' < t'$. □

Answer

(a) K is an intersection of closed sets. (b) Every point of K is approached by endpoints of shrinking intervals, so K is perfect with no intervals. (c) Binary addresses (left/right piece at each split) give an order-preserving bijection between any bounded Cantor set and K.

Exercise 4.7

Box-counting dimension of (a) the middle-$\frac{1}{5}$ Cantor set, (b) the Sierpinski carpet.

(a) Remove the middle fifth $(\frac{2}{5}, \frac{3}{5})$ of $[0,1]$. Two intervals of length $\frac{2}{5}$ remain, and the process repeats. After n steps there are 2^n intervals of length $(\frac{2}{5})^n$, so $N\big((\frac{2}{5})^n\big) \approx 2^n$ and

$$\text{boxdim} = \frac{\ln 2}{\ln(5/2)} \approx 0.7565.$$

(b) Divide the unit square into 9 squares of side $\frac{1}{3}$ and remove the middle one. Eight squares remain, and each is treated the same way. After n steps there are 8^n squares of side 3^{-n}. Boundaries of removed squares belong to the carpet, but this changes the box count only by a constant factor. So $N(3^{-n}) \approx 8^n$ and

$$\text{boxdim} = \frac{\ln 8}{\ln 3} = \frac{3\ln 2}{\ln 3} \approx 1.893.$$

Answer

(a) $\dfrac{\ln 2}{\ln(5/2)} \approx 0.757.$ (b) $\dfrac{\ln 8}{\ln 3} \approx 1.893.$

Exercise 4.8

Box-counting dimension of the set E of endpoints of the removed intervals of K.

$E = \{\frac{1}{3}, \frac{2}{3}, \frac{1}{9}, \frac{2}{9}, \frac{7}{9}, \frac{8}{9}, \dots\}$ is countable, yet its dimension is as large as that of K.

Upper bound. $E \subset K$, so $N_E(3^{-n}) \le N_K(3^{-n}) \le 2 \cdot 2^n$.

Lower bound. Each of the 2^n intervals of K_n contains an endpoint that lies in E (for $n \ge 1$). Distinct intervals of K_n are separated by gaps of length at least 3^{-n}, so a box of side 3^{-n} can meet at most 2 of them (meeting three would force the middle one, of length 3^{-n}, plus parts of two gaps, inside the box). Hence $N_E(3^{-n}) \ge 2^{n-1}$.

So $N_E(3^{-n})$ is within constant factors of 2^n, and $\text{boxdim}\, E = \dfrac{\ln 2}{\ln 3}$.

Answer

$\text{boxdim}(E) = \dfrac{\ln 2}{\ln 3} \approx 0.631$, the same as for K itself.

Exercise 4.9

Sierpinski-gasket variant: remove the upper-right quarter of the unit square, then of each remaining quarter, and so on. (a) Find the dimension. (b) Describe the gasket in binary.

(a) Each step keeps 3 of 4 quarters. After n steps there are 3^n squares of side 2^{-n}. So $N(2^{-n}) \approx 3^n$ and

$$\text{boxdim} = \frac{\ln 3}{\ln 2} \approx 1.585.$$

(b) Write $x = 0.x_1x_2\ldots_2$ and $y = 0.y_1y_2\ldots_2$. The first bits (x_1, y_1) tell which quarter of the square the point is in: $(0,0)$ lower left, $(1,0)$ lower right, $(0,1)$ upper left, $(1,1)$ upper right. Removing the upper-right quarter means *excluding* $(x_1, y_1) = (1,1)$. Inside each surviving quarter the picture is a scaled copy, and the second bits (x_2, y_2) play the same role, and so on. After all steps, the gasket is the set of (x, y) for which no n has $x_n = y_n = 1$ (for at least one choice of binary expansion when a point has two).

Answer

(a) $\dfrac{\ln 3}{\ln 2} \approx 1.585$. (b) The gasket consists of all (x, y) whose binary digits satisfy $(x_n, y_n) \neq (1,1)$ for every n.

Exercise 4.10

For bounded $A, B \subset \mathbb{R}$, show $\text{boxdim}(A \times B) = \text{boxdim}(A) + \text{boxdim}(B)$ *(assuming both dimensions exist).*

Use a grid of ε-boxes. A square $[i\varepsilon, (i+1)\varepsilon] \times [j\varepsilon, (j+1)\varepsilon]$ meets $A \times B$ exactly when the interval $[i\varepsilon, (i+1)\varepsilon]$ meets A *and* $[j\varepsilon, (j+1)\varepsilon]$ meets B. So the number of squares is the product of the interval counts:

$$N_{A\times B}(\varepsilon) = N_A(\varepsilon)\, N_B(\varepsilon).$$

Taking logs and dividing by $\ln(1/\varepsilon)$:

$$\frac{\ln N_{A\times B}(\varepsilon)}{\ln(1/\varepsilon)} = \frac{\ln N_A(\varepsilon)}{\ln(1/\varepsilon)} + \frac{\ln N_B(\varepsilon)}{\ln(1/\varepsilon)} \xrightarrow{\varepsilon\to 0} \text{boxdim}\, A + \text{boxdim}\, B. \qquad \square$$

Answer

$N_{A\times B}(\varepsilon) = N_A(\varepsilon) N_B(\varepsilon)$, so the dimensions add.

Exercise 4.11

True or false: the box-counting dimension of (a) a finite union, (b) a bounded countable union, is the maximum of the dimensions.

(a) True. For two sets A, B we have $\max(N_A, N_B) \le N_{A\cup B} \le N_A + N_B \le 2\max(N_A, N_B)$. Take logs and divide by $\ln(1/\varepsilon)$; the extra $\ln 2/\ln(1/\varepsilon) \to 0$. So $\text{boxdim}(A\cup B) = \lim \dfrac{\max(\ln N_A, \ln N_B)}{\ln(1/\varepsilon)} = \max(\text{boxdim}\, A, \text{boxdim}\, B)$. Induction extends this to any finite union.

(b) False. Take $S = \mathbb{Q} \cap [0,1]$. It is a countable union of single points, each of dimension 0. But every box of side ε in $[0,1]$ contains a rational, so $N(\varepsilon) \approx 1/\varepsilon$ and $\operatorname{boxdim} S = 1 \neq 0$.

Answer

(a) True. (b) False; counterexample: $\mathbb{Q} \cap [0,1]$ has dimension 1, although each point has dimension 0.

Exercise 4.12

(a) $S = \{0\} \cup \{1, \frac{1}{2}, \frac{1}{3}, \dots\}$ *has* $\operatorname{boxdim} = \frac{1}{2}$. *(b)* $S = \{0\} \cup \{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots\}$ *has* $\operatorname{boxdim} = 0$.

(a) The gap between $\frac{1}{k}$ and $\frac{1}{k+1}$ is $\frac{1}{k(k+1)}$. Fix n and take $\varepsilon_n = \dfrac{1}{n(n+1)} \approx n^{-2}$.

- The points $1, \frac{1}{2}, \dots, \frac{1}{n}$ are more than ε_n apart (smallest gap $\frac{1}{(n-1)n} > \varepsilon_n$), so each needs its own box: at least n boxes.
- The remaining points (and 0) lie in $[0, \frac{1}{n+1}]$. This has length $\frac{1}{n+1} = n\varepsilon_n$, so about n more boxes cover it.

Hence $n \leq N(\varepsilon_n) \leq 2n+1$, and

$$\frac{\ln N(\varepsilon_n)}{\ln(1/\varepsilon_n)} \approx \frac{\ln n}{\ln n(n+1)} \to \frac{1}{2}.$$

For ε between ε_{n+1} and ε_n, $N(\varepsilon)$ is squeezed between $N(\varepsilon_n)$ and $N(\varepsilon_{n+1})$, and $\ln(1/\varepsilon_{n+1})/\ln(1/\varepsilon_n) \to 1$, so the limit is the same: $\operatorname{boxdim} S = \frac{1}{2}$.

(b) Given ε, pick m with $2^{-m} \leq \varepsilon < 2^{-m+1}$. All points 0 and 2^{-k} with $k \geq m$ lie in $[0, 2^{-m}]$, which has length $\leq \varepsilon$ and needs at most 2 boxes. The points 2^{-k} with $k < m$ are only m points, so at most m boxes. So $N(\varepsilon) \leq m + 2 \leq \log_2(1/\varepsilon) + 3$ and

$$\frac{\ln N(\varepsilon)}{\ln(1/\varepsilon)} \leq \frac{\ln\big(\log_2(1/\varepsilon) + 3\big)}{\ln(1/\varepsilon)} \to 0.$$

Answer

(a) $N(\varepsilon) \approx 2\varepsilon^{-1/2}$, so $\operatorname{boxdim} = \frac{1}{2}$. (b) $N(\varepsilon)$ grows only like $\log(1/\varepsilon)$, so $\operatorname{boxdim} = 0$.

Exercise 4.13

Generalize: find $\operatorname{boxdim} S$ *for (a)* $S = \{0\} \cup \{n^{-p} : n \geq 1\}$, $p > 0$; *(b)* $S = \{0\} \cup \{p^{-n} : n \geq 0\}$, $p > 1$.

(a) Let $g(k) = k^{-p} - (k+1)^{-p}$. By the mean value theorem, $p(k+1)^{-(p+1)} \leq g(k) \leq pk^{-(p+1)}$. Given ε, let $n \approx (p/\varepsilon)^{1/(p+1)}$, so that $g(n) \approx \varepsilon$ (hence $n \asymp \varepsilon^{-1/(p+1)}$).

- For $k \lesssim n$ the gaps $g(k)$ are at least about ε, so no box contains more than 2 of these points: at least about $n/2$ boxes.

- Points with index $\geq n$ lie in $[0, n^{-p}]$, which needs about $n^{-p}/\varepsilon \approx n/p$ boxes, and the first n points need at most n boxes.

So $c_1\varepsilon^{-1/(p+1)} \leq N(\varepsilon) \leq c_2\varepsilon^{-1/(p+1)}$ and $\operatorname{boxdim} S = \dfrac{1}{p+1}$. (Check: $p = 1$ gives $\frac{1}{2}$, matching 4.12(a).)

(b) Same argument as 4.12(b). Choose m with $p^{-m} \leq \varepsilon < p^{-m+1}$. The points $0, p^{-k}$ with $k \geq m$ sit in an interval of length $\leq \varepsilon$ (at most 2 boxes), and the $m \approx \log_p(1/\varepsilon)$ points p^{-k}, $k < m$, need at most one box each. So $N(\varepsilon) = O(\log(1/\varepsilon))$ and $\operatorname{boxdim} S = 0$.

Answer

(a) $\dfrac{1}{1+p}$. (b) 0.

Exercise 4.14

Estimate the box-counting dimension from the table of box sizes $2^{-2}, \ldots, 2^{-9}$ *and boxes hit* 111, 327, 939, 2702, 7839, 22229, 62566, 178040.

If $N(\varepsilon) \approx C\varepsilon^{-d}$, then $\ln N = d\ln(1/\varepsilon) + \ln C$, so d is the **slope** of $\ln N$ against $\ln(1/\varepsilon)$. Here $\varepsilon = 2^{-k}$, so $\ln(1/\varepsilon) = k\ln 2$.

Step-by-step slopes. Between consecutive box sizes, $d \approx \log_2 \dfrac{N(2^{-k-1})}{N(2^{-k})}$:

$$1.56,\ 1.52,\ 1.52,\ 1.54,\ 1.50,\ 1.49,\ 1.51.$$

Least-squares line through all eight points: slope ≈ 1.520 with standard error ≈ 0.004 (so roughly ± 0.009 at 95% confidence). Using only the four finest scales gives slope ≈ 1.50.

How accurate is it? The step-by-step slopes drift from 1.56 down to 1.50, so the data are not perfectly a straight line. This systematic effect is larger than the regression's standard error, which in any case assumes independent data points, while here all points come from one orbit. A fair summary is $d \approx 1.5$, with an uncertainty of a few hundredths rather than a few thousandths.

Answer

boxdim ≈ 1.5 (least-squares slope 1.52 ± 0.01; finest scales give about 1.50).

Challenge 4: Fractal Basin Boundaries and the Uncertainty Exponent

The map is

$$F(x) = \begin{cases} 5x + 4, & x \leq -0.4, \\ -5x, & -0.4 \leq x \leq 0.4, \\ 5x - 4, & x \geq 0.4. \end{cases}$$

Let $B_{+\infty}$, $B_{-\infty}$ be the basins of $\pm\infty$, and J the set of points whose orbits stay in $[-1, 1]$ forever.

Key computation. Cut $[-1,1]$ into five pieces of length 0.4:

$$I_1 = [-1,-0.6],\ I_2 = [-0.6,-0.2],\ I_3 = [-0.2,0.2],\ I_4 = [0.2,0.6],\ I_5 = [0.6,1].$$

Evaluating F at the cut points gives:

Piece	Image under F	Meaning
I_1	$[-1,1]$ (slope $+5$)	stays, onto
I_2	$[1,2]$	escapes upward
I_3	$[-1,1]$ (slope -5)	stays, onto
I_4	$[-2,-1]$	escapes downward
I_5	$[-1,1]$ (slope $+5$)	stays, onto

For $x > 1$: $F(x) = 5x - 4 > x$, so the orbit increases to $+\infty$. For $x < -1$: $F(x) = 5x + 4 < x$, so the orbit decreases to $-\infty$.

Step 1: describe $B_{+\infty}$, $B_{-\infty}$, J

Points leave $[-1,1]$ only by landing in the interior of I_2 (exit up) or I_4 (exit down). Therefore:

- $B_{+\infty} = (1,\infty) \cup$ all $x \in [-1,1]$ such that, for some $n \geq 0$, $F^k(x) \in I_1 \cup I_3 \cup I_5$ for $k < n$ and $F^n(x) \in (-0.6,-0.2)$. In words: orbits whose first exit from $[-1,1]$ is through the top.
- $B_{-\infty} = (-\infty,-1) \cup$ all x whose first exit is through $(0.2,0.6)$.
- $J = \bigcap_n J_n$, where $J_n = \{x : F^k(x) \in [-1,1],\ 0 \leq k \leq n\}$. Since I_1, I_3, I_5 each map linearly (slope ± 5) onto $[-1,1]$, J_{n+1} is obtained by keeping the first, third and fifth fifths of every interval of J_n. So J_n consists of 3^n closed intervals of length $2 \cdot 5^{-n}$, and J is a Cantor set.

Step 2: $\operatorname{boxdim}(J) = \ln 3 / \ln 5$

Put $\varepsilon_n = 2 \cdot 5^{-n}$. The 3^n intervals of J_n cover J, and each meets at most 2 grid boxes, so $N(\varepsilon_n) \leq 2 \cdot 3^n$. Conversely, each interval contains points of J (its endpoints), and different intervals are separated by gaps of length at least ε_n, so a box of side ε_n meets at most 2 of them. So $N(\varepsilon_n) \geq 3^n/2$. Therefore

$$\operatorname{boxdim} J = \lim_{n\to\infty} \frac{\ln(3^n \cdot \text{const})}{\ln(5^n/2)} = \frac{\ln 3}{\ln 5} \approx 0.683.$$

Step 3: the basin is open

Let $x \in B_{+\infty}$. Its orbit tends to $+\infty$, so $F^n(x) > 1$ for some n. Since F^n is continuous, $F^n(y) > 1$ for all y in some neighborhood $N_\varepsilon(x)$. Once an orbit is above 1 it increases to $+\infty$, so $N_\varepsilon(x) \subset B_{+\infty}$. The same argument works for $B_{-\infty}$.

Step 4: every point of J touches both basins

Let $y \in J$ and $\varepsilon > 0$. Choose n with $2 \cdot 5^{-n} < \varepsilon$. Then y lies in an interval I of J_n of length $< \varepsilon$, so $I \subset N_\varepsilon(y)$. The map F^n sends I linearly onto $[-1, 1]$. Pick $z_1, z_2 \in I$ with $F^n(z_1) = 0.4$ and $F^n(z_2) = -0.4$. Then $F^{n+1}(z_1) = F(0.4) = -2$ and $F^{n+1}(z_2) = F(-0.4) = 2$. Both exit $[-1, 1]$, so $z_1 \in B_{-\infty}$ and $z_2 \in B_{+\infty}$. Thus $N_\varepsilon(y)$ contains points of both basins.

Step 5: $\varepsilon N(\varepsilon) \le L(\varepsilon) \le 3\varepsilon N(\varepsilon)$

An ε-uncertain point is a point within ε of J. Let $N(\varepsilon)$ be the number of grid intervals of length ε that meet J (the box-counting number of J), and $L(\varepsilon)$ the total length of the ε-uncertain set U_ε.

- *Lower bound.* Each of the $N(\varepsilon)$ grid intervals contains a point of J, so all its points are within ε of J. These intervals do not overlap, so $L(\varepsilon) \ge N(\varepsilon)\,\varepsilon$.
- *Upper bound.* If u is within ε of $j \in J$, and j lies in grid interval A_k, then $u \in A_{k-1} \cup A_k \cup A_{k+1}$. So U_ε is covered by $3N(\varepsilon)$ grid intervals, giving $L(\varepsilon) \le 3\varepsilon N(\varepsilon)$.

Step 6: $\lim_{\varepsilon \to 0} \dfrac{\ln L(\varepsilon)}{\ln \varepsilon} = p = 1 - \mathrm{boxdim}(J)$

Take logs in Step 5: $\ln \varepsilon + \ln N \le \ln L \le \ln 3 + \ln \varepsilon + \ln N$. Dividing by $\ln \varepsilon < 0$ *reverses* the inequalities:

$$1 + \frac{\ln 3}{\ln \varepsilon} + \frac{\ln N(\varepsilon)}{\ln \varepsilon} \;\le\; \frac{\ln L(\varepsilon)}{\ln \varepsilon} \;\le\; 1 + \frac{\ln N(\varepsilon)}{\ln \varepsilon}.$$

As $\varepsilon \to 0$, $\ln 3 / \ln \varepsilon \to 0$ and $\dfrac{\ln N(\varepsilon)}{\ln \varepsilon} = -\dfrac{\ln N(\varepsilon)}{\ln(1/\varepsilon)} \to -\,\mathrm{boxdim}\, J$. Both bounds tend to $1 - \mathrm{boxdim}\, J$, so the middle does too. In short, $L(\varepsilon) \sim \varepsilon^p$ with $p = 1 - \mathrm{boxdim}(J)$.

Step 7: the numerical example

If $\mathrm{boxdim}\, J = 0.9$ then $p = 0.1$ and $L(\varepsilon) \approx C\varepsilon^{0.1}$. To cut L by a factor of 2 we need $(\varepsilon'/\varepsilon)^{0.1} = \frac{1}{2}$, i.e.

$$\frac{\varepsilon'}{\varepsilon} = 2^{-10} = \frac{1}{1024}.$$

The measurement accuracy must improve by a factor of about 1000 just to halve the uncertain region. That is the practical cost of a fractal basin boundary.

Step 8: two-dimensional version

Now $J \subset \mathbb{R}^2$ is a bounded fractal basin boundary, and each point of J lies on the boundary of at least two basins. This hypothesis means that every point within ε of J is within ε of points of two different basins, so the ε-uncertain set U_ε is exactly the ε-neighborhood of J. Let $A(\varepsilon)$ be its area and $N(\varepsilon)$ the number of grid squares of side ε meeting J.

- *Upper bound.* A point within ε of $j \in J$ lies in the square of j or one of its 8 neighbors. So $A(\varepsilon) \le 9\varepsilon^2 N(\varepsilon)$.

- *Lower bound.* Use squares of side $\varepsilon/2$. Any such square meeting J lies entirely within distance $\sqrt{2}\,\varepsilon/2 < \varepsilon$ of J, so $A(\varepsilon) \geq \dfrac{\varepsilon^2}{4} N(\varepsilon/2)$.

Take logs, divide by $\ln \varepsilon < 0$ (reversing inequalities), and let $\varepsilon \to 0$, using $\ln N(\varepsilon)/\ln \varepsilon \to -d$ and $\ln N(\varepsilon/2)/\ln \varepsilon \to -d$, where $d = \operatorname{boxdim} J$:

$$\lim_{\varepsilon \to 0} \frac{\ln A(\varepsilon)}{\ln \varepsilon} = 2 - d = p.$$

So $A(\varepsilon) \sim \varepsilon^p$ with $p = 2 - \operatorname{boxdim}(J)$. □

Summary of answers

Exercise	Result
4.1	$f(x) = 3^n(x-a)$ matches $K \cap [a, a+3^{-n}]$ with K (note on (b) misprint)
4.2	removed length 1; base-4 digits $\{0,3\}$; $\frac{1}{5}, \frac{17}{21} \in K(4)$; boxdim $= \frac{1}{2}$; $a = 4$
4.3	rationals↔rationals; periodic points rational; irrationals give uncountably many nonperiodic points
4.4	cover q_n by length $\varepsilon/2^n$
4.5	$\sum a_i 3^{-i} \mapsto \sum (a_i/2) 2^{-i}$
4.6	closed, perfect, no intervals; order-preserving bijection via binary addresses
4.7	(a) $\ln 2/\ln(5/2)$; (b) $\ln 8/\ln 3$
4.8	$\ln 2/\ln 3$
4.9	(a) $\ln 3/\ln 2$; (b) bits $(x_n, y_n) \neq (1,1)$ for all n
4.10	dimensions add: $N_{A\times B} = N_A N_B$
4.11	(a) true; (b) false ($\mathbb{Q} \cap [0,1]$)
4.12	(a) $\frac{1}{2}$; (b) 0
4.13	(a) $\frac{1}{1+p}$; (b) 0
4.14	boxdim ≈ 1.5
Challenge 4	$\operatorname{boxdim} J = \ln 3/\ln 5$; $L(\varepsilon) \sim \varepsilon^{1-\operatorname{boxdim} J}$; need $\approx 1024\times$ better accuracy in Step 7; $A(\varepsilon) \sim \varepsilon^{2-\operatorname{boxdim} J}$